

UP Board 2020

Grade 12

Mathematics

Time: 3 Hours 15 Minutes

Max. Marks: 100

Note: First 15 minutes are allotted for the candidates to read the question paper.

Instructions:

- i) There are in all nine question in this question paper.
- ii) All questions are compulsory.
- iii) In the beginning of each question, the number of parts to be attempted are clearly mentioned.
- iv) Marks allotted to the questions are indicated against them.
- v) Start solving from the first question and proceed to solve till the last one.
- vi) Do not waste your time over a question you cannot solve.

1. Attempt all the parts :

a) Suppose that the function $f: R \rightarrow R$ is defined by $f(x) = x^2$. Then

f is

- i) one-one onto
- ii) many-one onto

iii) one-one, but not onto

iv) neither one-one nor onto.

1

b) The value of $\cos \theta \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} + \sin \theta \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$

i) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

ii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

iii) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

iv) none of these.

1

c) The value of integral $\int \sqrt{1 + \sin 2x} dx$

i) $\sin x + \cos x + c$

ii) $\sin x - \cos x + c$

iii) $\cos x - \sin x + c$

iv) $-\sin x - \cos x + c$

1

d) The degree of the differential equation $\frac{d^2 y}{dx^2} = \sqrt[4]{x + \left(\frac{dy}{dx}\right)^2}$

i) 4

ii) 3

iii) 2

iv) 1

1

c) If vector $5\hat{i} - \lambda\hat{j} + 2\hat{k}$ and $2\hat{i} + 3\hat{j} + 4\hat{k}$ are perpendicular to each other, then value of λ is

- i) 3
- ii) 4
- iii) 6
- iv) 0

2. Attempt all the parts:

a) Find the principal value of $\sin^{-1}\left(\sin\frac{7\pi}{4}\right)$ 1

b) Function $f: R \rightarrow R$ is such that $f(x) = x^2 + 2$, then find $f^{-1}(18)$ 1

c) If A and B are two events such that $P(A) = 0.4$, $P(B) = 0.8$ and $p(B/A) = 0.6$, then find $P(A \cap B)$. 1

d) Find the differential equation, related to equation $y = Aex + B$, where A, B are constants.

e) Show that for function $f(x) = \begin{cases} x^3 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$ is not continuous. 1

3. Attempt all the parts:

a) If $y = \cot^{-1}\left(\frac{\sqrt{1+x^2}+1}{x}\right)$ then find the value of $\frac{dy}{dx}$ 2

b) Find the angle between the vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $3\hat{i} + 3\hat{j} + 4\hat{k}$ 2

c) Under the following restriction

$$x + y \leq 8$$

$$3x + 5y \geq 15$$

$$x \geq 0, y \geq 0$$

Find the minimum value of $Z = x + 3y$. 2

d) Find the value of p if the lines $\frac{1-x}{x} = \frac{7y-14}{2p} = \frac{z-3}{2}$ and

$\frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$ are mutually. 2

4. Attempt all the parts :

a) In a class 40% students read mathematics, 20% . read biology and 10% read both mathematics and biology. One student is selected at random. Find the probability that

i) he reads mathematics, when it is known that he reads biology

ii) he reads biology when it is known that he reads

mathematics. 1 + 1

b) Find the value of $\int \frac{\sin(x-a)}{\sin(x+a)} dx$, where a is constant. 2

c) Find the area of that parallelogram whose diagonals are

$\vec{d}_1 = 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{d}_2 = \hat{i} - 3\hat{j} + 2\hat{k}$ 2

d) Verify Rolle's theorem for the function $f(x) = \cos 2x$ in the interval $[0, \pi]$ 2

5. Attempt any five parts of the following

a) If $\cos^{-1} \frac{x}{a} + \cos^{-1} \frac{y}{b} = \theta$, then prove that $\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \theta + \frac{y^2}{b^2} = \sin^2 \theta$ 5

b) Prove that

$$\begin{vmatrix} a+b & b & c \\ b+c & c & a \\ c+a & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3$$

5

c) prove that $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{1 + \sqrt{\cot x}} dx = \frac{\pi}{4}$ 5

d) If curve $y = x^3$ and $xy = k$ cut each other perpendicular, then prove that $3k = 1$ 5

e) Find the shortest distance between the lines

$$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}) \quad \text{and} \quad \vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$$

5

f) Solve the differential equation $x \frac{dy}{dx} = y - x \cos^2 \left(\frac{y}{x} \right)$ 5

6. Attempt any five parts of the following:

a) show that the function $f(x) = |x - 2| = \begin{cases} x - 2, & x \geq 2 \\ -(x - 2), & x < 2 \end{cases}$

at $x = 2$ is Continuous but not differentiable 5

b) Solve the following linear programming problem by

graphical method, under the following Constraints:

$$x + 3y \leq 60$$

$$x + y \geq 10$$

$$x \leq y$$

$$x \geq 0, y \geq 0$$

Find the minimum and maximum values of $Z = 3x + 9y$ 5

c) Find the area enclosed in the first quadrant of circle $x^2 + y^2 = 4$ and straight line $y = x$ and axis x . 5

d) In a bag there are 5 red, 4 black and 3 white balls. If one by one three balls are taken out and not again returned to the bag, then find the probability that all the three balls drawn are red. 5

e) Find the perpendicular unit vector $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$ and find the sine of the angle between them 5

f) Solve the equation

$$\sec^{-1} \frac{x}{a} - \sec^{-1} \frac{x}{b} = \sec^{-1} b - \sec^{-1} a$$
5

7. Attempt any one part of the following

a) Solve the following system of equations by matrix method:

$$3x - 2y + 3z = 8$$

$$2x + y - z = 1$$

$$4x - 3y + 2z = 4$$
8

b) i) If area of the triangle, Whose vertices are (2, -6), (5, 4) (k, 4) is 35 square unit, then find the value of k. 3

ii) If $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$,

$$\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k} \text{ and}$$

$\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$, then find a vector $\vec{d}$ which is

Perpendicular to vectors $\vec{a}$ and $\vec{b}$ both and $\vec{c} \cdot \vec{d} = 15$. **5**

8. Attempt any one part of the following:

a) Find the value of $\int \frac{x^2 + 1}{x^2 - 5x + 6} dx$ **8**

b) i) Find the minimum value of the function x^x . **4**

ii) From a pack of 52 playing cards, two are drawn one by one. After drawing they are not returned to the pack again, then find the probability of both cards to be betel. **4**

9. Attempt any one part of the following:

a) Find the value of $\int_a^b x^2 dx$ with the help of definite as the limit of a sum. **8**

b) Find the general solution of the differential equation

$$\frac{dy}{dx} + 2y = \sin x$$
 8